

HDOC Day 37

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Batch - 15

Algorithm

Main Program

1. Start.
2. Display the menu.
3. Read the user's choice.
4. Use `switch` to call the corresponding function.
5. For choice 1, call `toUppercase()`.
6. For choice 2, call `reverseString()`.
7. For choice 3, call `checkPalindrome()`.
8. For choice 4, exit.
9. For any other choice, display an error message.
10. Repeat until choice 4 is selected.
11. Stop.

Operation 1 – Lowercase to Uppercase

1. Read the string.
2. Traverse characters until `'\0'`.
3. If a character lies between `'a'` and `'z'`, convert it using:
`str[i] = str[i] - ('a' - 'A')`
4. Display the uppercase string.

Operation 2 – Reverse String

1. Read the string.
2. Determine its length by traversing until `'\0'`.
3. Swap the first character with the last, second with second-last, and so on.
4. Continue until half of the string has been processed.
5. Display the reversed string.

Operation 3 – Palindrome

1. Read the string.
2. Determine its length without `strlen()`.
3. Compare the first character with the last, second with second-last, etc.
4. If any pair differs, the string is not a palindrome.
5. Otherwise, it is a palindrome.

Test Case Table

Test	Choice	Input	Expected Output
1	1	hello world	HELLO WORLD
2	1	computer	COMPUTER
3	2	hello	olleh
4	2	programming	gnimmargorp
5	3	madam	Palindrome
6	3	level	Palindrome
7	3	hello	Not palindrome
8	5	—	Invalid choice
9	4	—	Exit

C Program

```
#include <stdio.h>
```

```
void toUppercase(void);
void reverseString(void);
void checkPalindrome(void);
```

```
int main()
{
    int choice;

    do
    {
        printf("\n===== STRING OPERATIONS =====\n");
        printf("1. Convert lowercase string to uppercase\n");
        printf("2. Reverse a string\n");
        printf("3. Check if a string is a palindrome\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                toUppercase();
                break;

            case 2:
                reverseString();
                break;

            case 3:
                checkPalindrome();
                break;
```

```

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice! Please enter 1 to 4.\n");
    }

    } while (choice != 4);

    return 0;
}

```

```

/* Function to convert lowercase string to uppercase */
void toUppercase(void)
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf(" %[^\\n]", str);

    for (i = 0; str[i] != '\\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - ('a' - 'A');
        }
    }

    printf("Uppercase string: %s\\n", str);
}

```

```

/* Function to reverse a string */
void reverseString(void)
{
    char str[100], temp;
    int i, length = 0;

    printf("Enter a string: ");
    scanf(" %[^\\n]", str);

    /* Find length without strlen() */
    while (str[length] != '\\0')
    {
        length++;
    }

    for (i = 0; i < length / 2; i++)
    {
        temp = str[i];
        str[i] = str[length - 1 - i];
        str[length - 1 - i] = temp;
    }
}

```

```

    printf("Reversed string: %s\n", str);
}

/* Function to check palindrome */
void checkPalindrome(void)
{
    char str[100];
    int i, length = 0;
    int palindrome = 1;

    printf("Enter a string: ");
    scanf(" %[^\\n]", str);

    /* Find length without strlen() */
    while (str[length] != '\\0')
    {
        length++;
    }

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            palindrome = 0;
            break;
        }
    }

    if (palindrome)
        printf("The string is a palindrome.\\n");
    else
        printf("The string is not a palindrome.\\n");
}

```

EXECUTION

```

[aaryanrajput@aaryans-macbook ~ % nano day37.c
[aaryanrajput@aaryans-macbook ~ % clang day37.c -o day37.c
[aaryanrajput@aaryans-macbook ~ % ./day37.c

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 1
Enter a lowercase string: hello
Uppercase string: HELLO

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 2
Enter a string: hello
Reversed string: olleh

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 3
Enter a string: naman
The string is a palindrome.

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 3
Enter a string: hello
The string is not a palindrome.

===== STRING OPERATIONS =====
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 4
Exiting program...
aaryanrajput@aaryans-macbook ~ % █

```